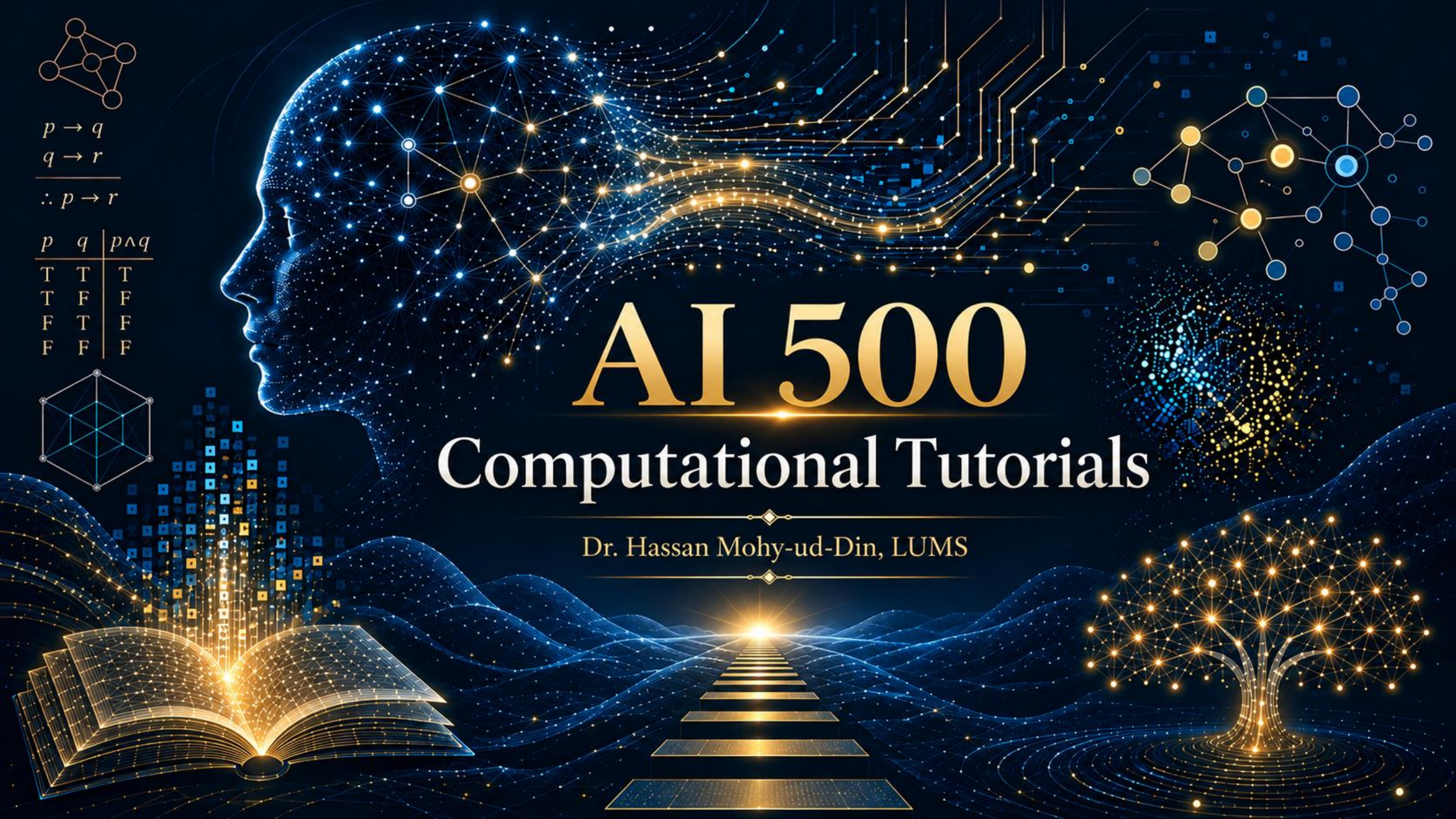

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AI 500
Computational Tutorials
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DISCLAIMER: The cover image was generated by ChatGPT 5.6 sol using the Syllabus document as the body of a prompt.

1 Problem Setup

Synthetic Multi-Hospital Clinical Deterioration Dataset

Problem Statement: Develop and evaluate a predictive system that estimates whether an adult patient will experience clinical deterioration within the next 12 hours, using only information that is available within 1 hour of hospital admission.

- **Unit of observation:** one hospital encounter
- **Unit of prediction:** deterioration risk for the current encounter
- **Prediction time:** 1 hour after admission
- **Prediction horizon:** subsequent 12 hours
- **Primary outcome:** deterioration_within_12h
- **Population:** synthetic adult patients admitted across **8 hospitals**
- **Dataset scale:** approximately **250,000 encounters** from **118,000 patients**
- **Time span:** 2021 – 2026

1 Problem Setup

Synthetic Multi-Hospital Clinical Deterioration Dataset

Problem Statement: Develop and evaluate a predictive system that estimates whether an adult patient will experience clinical deterioration within the next 12 hours, using only information that is available within 1 hour of hospital admission.

- **Available data domains:** demographics, comorbidities, vital signs, laboratory measurements, prior utilization, hospital/device characteristics, and timestamps
- **Realistic data challenges:** missing values, repeated patients, class imbalance, duplicates, inconsistent labels, outliers, mixed units, measurement bias, temporal shifts, and subgroup heterogeneity
- **Critical constraint:** only features genuinely available by the prediction time may be used
- **Leakage:** ICU transfer, rapid-response activation, discharge disposition, and length of stay

2 From Raw Data to Trustworthy Data

2.1 Data Acquisition from Heterogenous Sources

Computational Focus: Bring data into Python while preserving source, format, time, and provenance information.

- **Load tabular data from CSV** using pandas.read_csv().
- **Inspect schemas immediately after ingestion:** columns, data types, identifiers, and timestamps.
- **Record where** the data came from**, when** it was obtained**,** and **what** each source represents.
- **Keep acquisition separate from cleaning** so that the original input remains reproducible.

Key teaching point: Data acquisition is not merely file loading; it is the first step in establishing provenance and meaning.

2 From Raw Data to Trustworthy Data

2.2 Initial Dataset Inventory

Computational Focus: Establish what the dataset actually contains before changing any values.

- **Determine the number of** rows, columns, patients, encounters, hospitals, and years represented.
- **Inspect** column names, data types, memory use, identifiers, and candidate target variables.
- **Check** the observed time span and the number of unique entities.
- **Quantify** class prevalence for deterioration_within_12h.
- **Distinguish** the unit of observation from the number of independent patients.

Key teaching point: Before cleaning, know the dataset's scope, structure, units, target, and population.

2 From Raw Data to Trustworthy Data

2.3 Automated Data Profiling

Computational Focus: Generate a systematic first-pass summary of structure, completeness, uniqueness, and distributions.

- **Compute missing-value** percentages for every variable.
- **Count unique values** and identify high-cardinality or near-constant variables.
  - High-cardinality categorical features can increase model complexity and make encoding more difficult.
  - Near-constant columns often provide little information for machine learning models.
- **Summarize** numeric **distributions** and categorical frequencies.
- **Detect exact duplicate** rows and duplicated encounter identifiers.
- **Create** an HTML profile with **ydata-profiling** for rapid exploration.

Key teaching point: Profiling identifies where to investigate; it should not automatically decide how to clean.

2 From Raw Data to Trustworthy Data

2.4 Schema and Data Validation

Computational Focus: Translate domain expectations into explicit computational checks.

- **Verify** that required columns exist and identifiers satisfy expected uniqueness rules.
- **Check** allowable ranges for age, oxygen saturation, heart rate, temperature, and blood pressure.
- **Validate** categorical domains and binary outcome values.
- **Check logical and temporal constraints**, such as prediction time occurring after admission.
- **Use assertions or libraries** such as pandera to make validation reproducible.

Key teaching point: A validation rule should express a testable expectation about structure, logic, or domain plausibility.

2 From Raw Data to Trustworthy Data

2.5 Data-Quality Dimensions

Computational Focus: Measure data quality as several distinct properties rather than one overall score.

- **Completeness:** quantify missing cells, missing records, and missingness by subgroup.
- **Accuracy:** identify impossible values, unit errors, and known reference inconsistencies.
- **Consistency:** examine categorical labels, logical relationships, and cross-field agreement.
- **Timeliness:** calculate recording, arrival, and result-availability delays.
- **Representativeness and measurement:** compare hospital composition and device-specific distributions.

Key teaching point: A dataset can be complete yet biased, consistent yet unrepresentative, or accurate yet too late to use.

2 From Raw Data to Trustworthy Data

2.6 Missing-Data Investigation

Computational Focus: Study where, how, and for whom data are missing before selecting an imputation strategy.

- **Measure variable-level missingness:** what fraction of values are absent in each column?
- **Measure record-level missingness:** which encounters are unusually incomplete?
- **Inspect missingness patterns** across related laboratory variables.
- **Compare missingness** by hospital, rural/urban setting, and socioeconomic group.
- **Recognize informative missingness:** lactate is more often measured in clinically concerning patients.

Key teaching point: Missingness can itself encode the data-generating process; missing is not automatically random.

2.7 Missing-Value Investigation

Computational Focus: Compare alternative strategies and fit every learned imputation rule using training data only.

- **Deletion** may be reasonable only when the loss of information and potential bias are acceptable.
- **Use median or mode imputation** for simple robust baselines.
- **Compare groupwise imputation**, for example hospital-specific medians.
- **Demonstrate KNN** or model-based imputation for multivariable structure.
- **Add missingness indicators** when the fact that a value is absent may itself be informative.

Key teaching point: Imputation is a modeling decision, not clerical repair and it can introduce leakage if fitted globally.

2.8 Outlier and Anomaly Investigation

Computational Focus: Flag unusual observations, then determine whether they are errors, rare valid cases, or evidence of shift.

- **Use the IQR rule** and standardized scores as screening tools.
- **Apply physiological plausibility** checks to heart rate, temperature, and blood pressure.
- **Compare statistical outliers** with clearly impossible values.
- **Inspect** source, unit, device, hospital, and timestamp context before correction or exclusion.
- **Document** whether each anomaly is retained, corrected, winsorized, transformed, or removed.

Key teaching point: An outlier flag is a request for investigation; not an instruction to delete the observation.

2.9 Duplicate and Entity-Resolution Problems

Computational Focus: Separate duplicate records from legitimate repeated observations of the same entity.

- **Detect exact duplicate** rows using duplicated().
- **Check duplicated** encounter identifiers as a key-integrity problem.
- **Construct candidate keys** such as patient & hospital & admission time to find near duplicates.
- **Count repeated encounters** contributed by each patient.
- **Recognize** false merges and false splits as entity-resolution errors.

Key teaching point: Row count is not sample size when patients contribute repeated encounters.

2 From Raw Data to Trustworthy Data

2.10 Units, Labels, and Consistency Checks

Computational Focus: Harmonize representations while preserving enough metadata to understand the original values.

- **Standardize** inconsistent labels such as Female, F, and female.
- **Harmonize** smoking-status codes while retaining the raw field for auditability.
- **Convert** weight from pounds to kilograms using the explicit weight_unit field.
- **Flag** implausible values after conversion to reveal possible unit-label errors.
- **Use** data dictionaries to define units, valid ranges, precision, and transformation rules.

Key teaching point: A numerical value without a trustworthy unit and definition is not a trustworthy measurement.

2 From Raw Data to Trustworthy Data

2.11 Class Imbalance

Computational Focus: Show why high accuracy can coexist with failure to identify the minority class.

- **Measure** the prevalence of `deterioration_within_12h` before model fitting.
- **Fit** a majority-class `DummyClassifier` as a deliberately weak baseline.
- **Compare** accuracy with balanced accuracy, recall, precision, and F_1.
- **Demonstrate** random oversampling and SMOTE using imbalanced-learn.
- **Discuss** class weighting and threshold adjustment as alternatives to synthetic oversampling.

Key teaching point: Any resampling strategy must be applied only to training data or inside each training fold.

2.12 Split Before Preprocessing

Computational Focus: Protect validation and test data before estimating any data-dependent preprocessing parameter.

- **Create** training, validation, and final test partitions before imputation or scaling.
- **Preserve** outcome prevalence with stratification when appropriate.
- **Fit** medians, means, scaling parameters, encoders, feature selection, and PCA on training data only.
- **Apply** the fitted transformation unchanged to validation and test observations.
- **Keep** the final test set isolated from iterative model-development decisions.

Key teaching point: The test set must represent unseen evidence; not a source of preprocessing information.

2 From Raw Data to Trustworthy Data

2.13 Data Leakage Demonstration

Computational Focus: Deliberately compare a valid admission-time model with a model contaminated by future information.

- **Safe predictors** include demographics, admission-time vital signs, comorbidities, and valid laboratory results.
- **Leaked predictors** include ICU transfer, rapid-response activation, discharge disposition, and length of stay.
- **Use** `lab_result_available_timestamp` to determine whether a laboratory value truly existed by prediction time.
- **Fit** the same model twice (with and without leaked features) and compare apparent performance.
- **Ask of every feature:** could this exact information exist at the intended prediction time?

Key teaching point: Leakage produces impressive-looking evaluation results precisely because the model has been given information from the future.

2.14 Cross-Validation Appropriate to Data Structure

Computational Focus: Choose a resampling scheme that reflects the dependence and deployment structure of the data.

- **KFold** assumes rows can be partitioned independently.
- **StratifiedKFold** additionally preserves class proportions across folds.
- **GroupKFold** keeps all encounters from the same patient in one fold, preventing patient leakage.
- **TimeSeriesSplit** respects chronological ordering and evaluates future-like observations.
- **Hospital-level holdout** can test whether the model generalizes to a genuinely unseen institution.

Key teaching point: The correct cross-validation design depends on what “new data” will mean at deployment.